A1 Challenge BSW — Figures and Tables

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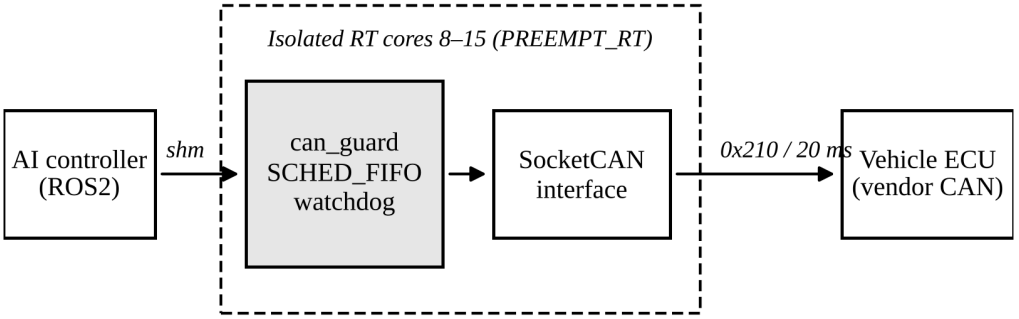


Fig. 1. BSW layer on a single main PC. *can_guard* runs outside ROS2 on isolated cores and is the only process that writes vehicle commands.

TABLE I
SUMMARY OF BSW VERIFICATION RESULTS

Metric	Result	Condition
Worst-case latency (30 min)	467 → 27 µs	cyclictest
Events > 400 µs (8 h)	231 → 26	cyclictest
Idle power	35.0 → 14.2 W	RAPL
Send-loop latency, max (8 h)	44.6 µs	SIL, virtual CAN
Spurious state transitions	0 / 2.88 M cycles	SIL, 8 h
Vehicle frames validated	890,611 (0 errors)	lift test, 73 min

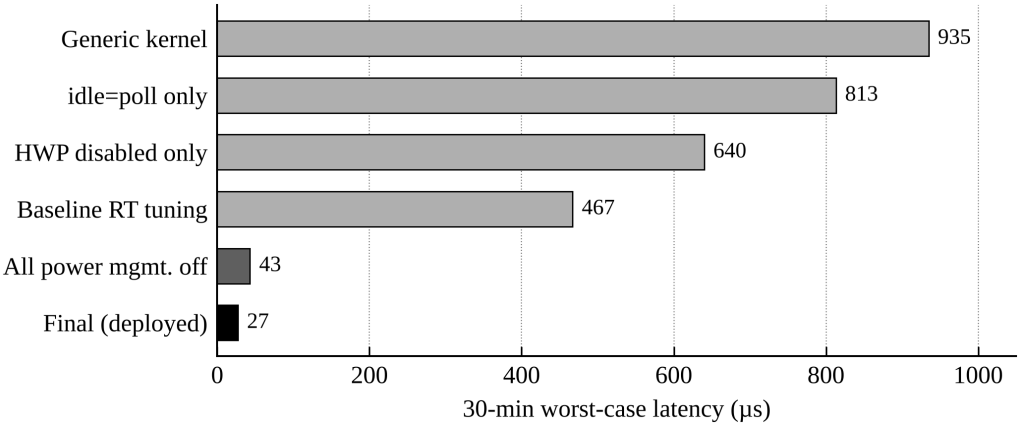


Fig. 2. 30-min worst-case latency for each kernel and power-management configuration (lower is better).

TABLE II
BASELINE RT TUNING VS. DEPLOYED CONFIGURATION

Metric	Baseline RT	Deployed
Worst-case latency, 30 min	467 µs	**27 µs**
Events > 400 µs, 8 h	231	**26**
Worst-case latency, 8 h	828 µs	920 µs
Idle power	35.0 W	**14.2 W**

Deployed: eco 800 + ondemand governor. The 8-h worst case did not decrease.

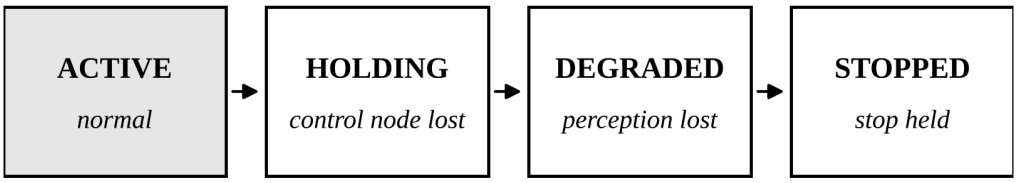


Fig. 3. Main states of *can_guard* (simplified). On a fault, steering is held, throttle is set to zero, and the brake is kept applied.

TABLE III
FAULT-INJECTION AND SOAK TESTS OF CAN_GUARD

Test	Condition	Result
P-1	Control node killed (SIGKILL)	HOLDING after 69 ms
P-2	Perception process killed	Period kept (20.00 ms), DEGRADED
Soak	8 h continuous run	2.88 M cycles, 0 transitions
Unit	Limits, real-frame replay	68 tests
Bugs	Found during development	6 fixed

SIL on a virtual CAN bus; latency excludes hardware delay.

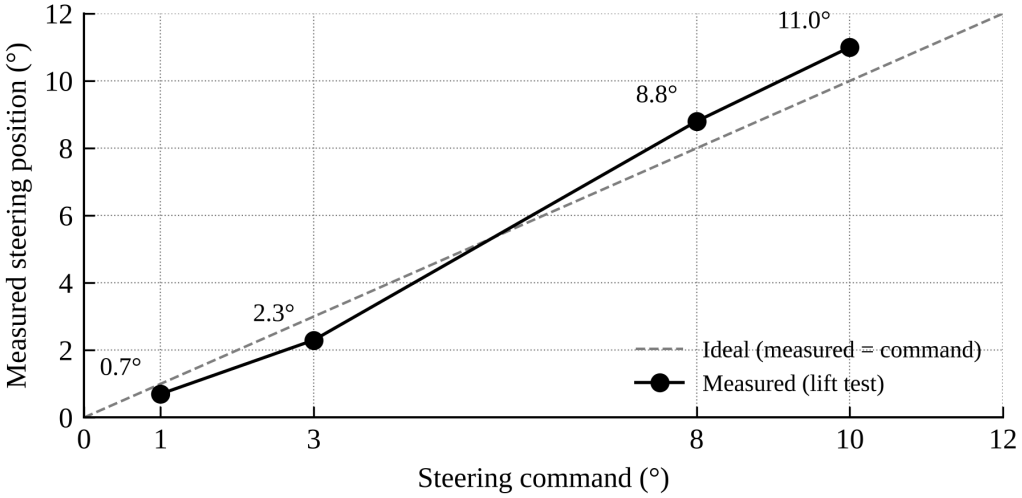


Fig. 4. Steering command versus measured steering position on the lift test. The data confirmed a scale of ×0.1, correcting the initial ×1 hypothesis.

TABLE IV
IDENTIFIED ACTUATOR AND TIMING CHARACTERISTICS

Item	Steering	Brake
Dead time	167–190 ms	56–112 ms
Time to 63 % of step	192–262 ms	–
End-to-end delay	≈ 0.23 s	≈ 0.15 s
Deadband / scale	±1.5°	position = cmd ÷ 10

Cycle: command 0x210 20.0 ms, vehicle status 21.0 ms → 0.42 s phase beat. 0.23 s ≈ 3.2 m at 50 km/h.